

Cut-Objective Predictive Mechanics

Boundary response geometry from composable region weights

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Abstract

This paper reconstructs a specific piece of phase geometry without taking a Newtonian state space, Lagrangian action, Hamiltonian phase space, or Hamilton-Jacobi principal function as primitive. The primitive object is the total weight of histories in a bounded region compatible with prescribed boundary data, denoted $K_R(\varphi_{\partial R})$, where R is the region, ∂R is the boundary of R , and $\varphi_{\partial R}$ denotes the boundary data. If R is split into subregions R_1 and R_2 by an auxiliary internal interface Σ , a family of region weights is cut-objective when the whole-region weight is recovered by the gluing relation $K_R(\varphi_{\partial R}) = \int K_{R_1}(\varphi_{\partial R_1}) K_{R_2}(\varphi_{\partial R_2}) \mu_\Sigma(d\varphi_\Sigma)$, where φ_Σ ranges over admissible data on Σ and μ_Σ is the gluing measure. The contribution is a conditional reconstruction: for a positive family of such weights K_R^ε , where $\varepsilon > 0$ is a sharpness parameter such as a noise strength, assume the sharp generator W_R exists as the leading exponential cost, $W_R(\varphi_{\partial R}) = -\lim_{\varepsilon \rightarrow 0} \varepsilon \log K_R^\varepsilon(\varphi_{\partial R})$. Under explicit Laplace and regularity hypotheses, exact gluing becomes a variational rule: the cost of the whole region is the infimum, over internal interface data, of the two subregion costs plus any exponential rate contributed by the gluing measure. Boundary derivatives of W_R define response covectors. At a smooth interior optimum on an internal cut, stationarity relates the oriented responses on the two sides of the cut; they are equal when the gluing measure contributes no first-order interface-rate correction at the sharp scale. For a smooth finite-dimensional boundary-configuration manifold, a twice differentiable endpoint version of W_R yields a Lagrangian endpoint response relation in the product of incoming and outgoing cotangent bundles; if the mixed Hessian is nondegenerate, this relation is locally the graph of a symplectic map. The novelty is in the route by which the generator W_R and its response covectors arise from cut-objective predictive composition. Brownian and Ornstein-Uhlenbeck transition kernels illustrate the construction.

1 introduction

Most formulations of mechanics begin only after a kinematic arena has already been fixed. Newtonian mechanics starts from positions and velocities. Hamiltonian mechanics starts from a phase space with a symplectic form [3]. Variational mechanics starts from a configuration space and an action functional, while Hamilton-Jacobi theory starts from a principal function whose endpoint derivatives generate momenta and canonical transformations that preserve the symplectic form [3, 24]. By contrast, stochastic and statistical mechanics provide a more compositional starting point: kernels compose by integrating over intermediate states, and large-deviation limits turn probabilities into rate functions [16, 13, 29]. Similarly, path integrals and general-boundary formulations assign amplitudes to regions and compose them by summing over boundary data [14, 25, 26].

We ask what can be reconstructed if one starts from the compositional side rather than from a preselected configuration space, phase space, or field space. More precisely, we ask when the variables used to glue regions across an interface acquire conjugate response covectors. The word “predictive”

is used in this paper in a specific boundary-compositional sense: boundary data are predictive when they are sufficient to reproduce whole-region boundary weights under admissible gluing. No theorem about optimal forecasting, causal-state minimality, or Bayesian filtering is assumed.

The reconstruction proved below is restricted to a regular finite-dimensional setting with positive composable weights. This covers the clean endpoint and finite-interface case and gives a controlled model of the mechanism; extension to infinite-dimensional field-theoretic boundary spaces would require additional analytic assumptions and is not claimed here. The construction has three steps. Exact gluing of region weights, together with a sharp large-deviation limit, produces a differentiable generator W . Derivatives of W with respect to boundary or interface variables define conjugate response covectors. These boundary-response pairs then determine a Lagrangian response relation, with a symplectic map only under an additional nondegeneracy condition. We use the standard Hamilton-Jacobi and generating-function step only at the end. Thus, the contribution is to reconstruct familiar Hamilton-Jacobi response geometry from upstream compositional history weights in a sharp regime, rather than assuming it as mechanical input. The added structure is the gluing-sufficiency test for boundary variables; the resulting response variables are then derivatives of a sharp generator, and their symplectic behavior is standard generating-function geometry. The paper derives this response geometry, not equations of motion, a global Hamiltonian flow, or new empirical predictions.

A simple orientation example is Brownian motion. It displays the two operations used throughout the paper: composition of weights and passage to a sharp cost. Let $x, y, z \in \mathbb{R}^d$, let $T_1, T_2 > 0$ be elapsed times, and let $T = T_1 + T_2$. The small-noise Brownian transition density $K_T^\varepsilon(x, z)$ composes by integration over an intermediate position,

$$K_{T_1+T_2}^\varepsilon(x, z) = \int K_{T_1}^\varepsilon(x, y) K_{T_2}^\varepsilon(y, z) dy.$$

In the small-noise limit its exponent defines the cost $W_T(x, z) = |z - x|^2/(2T)$. The composition law becomes

$$W_{T_1+T_2}(x, z) = \inf_y [W_{T_1}(x, y) + W_{T_2}(y, z)],$$

where y is the intermediate position at elapsed time T_1 , the shared interface between the intervals $[0, T_1]$ and $[T_1, T_1 + T_2]$. If y_* is a smooth interior minimizer, stationarity equates the oriented boundary derivatives of the two costs at y_* . Thus the intermediate variable carries a conjugate covector. This example demonstrates the general derivation used in the paper: a phase-space-like response covector can arise from compositional weights and a sharp limit before any mechanical action, Hamiltonian, or phase space has been introduced.

In this paper a region is defined by its compositional role. A region R is a finite part of a theory's domain for which four ingredients are specified. First, there are interior histories or configurations $h_R \in \mathcal{H}_R$, meaning assignments of the theory's variables inside R : paths for a time interval, variables on internal sites for a lattice block, or field configurations in a spacetime subdomain. Thus a trajectory is the special case of an interior history for a time interval. Second, there is an interface ∂R separating the region from its exterior. Third, there are boundary data $\varphi_{\partial R} \in \mathcal{B}_{\partial R}$, where $\mathcal{B}_{\partial R}$ is the space of admissible data exposed on that interface. Fourth, there is a region weight $K_R : \mathcal{B}_{\partial R} \rightarrow [0, \infty)$. For a fixed boundary value $\varphi_{\partial R}$, this weight assigns the total weight of all interior histories or configurations compatible with that boundary value. In schematic continuum notation,

$$K_R(\varphi_{\partial R}) = \int_{\{h_R \in \mathcal{H}_R : h_R|_{\partial R} = \varphi_{\partial R}\}} w_R[h_R] \mathcal{D}h_R.$$

Here the domain of integration is the set of interior histories whose boundary restriction equals the prescribed boundary data, and $\mathcal{D}h_R$ denotes the corresponding formal measure over histories. In

finite or discrete models this notation becomes a sum over compatible assignments. Depending on context, K_R may be a probability density, an unnormalised statistical weight, a path-space likelihood, a large-deviation exponential weight, or a quantum amplitude. The positive-weight results apply directly to the probabilistic, statistical, likelihood, and large-deviation cases. Quantum amplitudes require separate stationary-phase assumptions; this paper uses only their single-branch sharp limit as an analogy to the positive case.

The key requirement is composability: R can be glued to another region by summing or integrating over shared interface data. As an example, a Markov time interval is a region in this compositional sense. Its interior histories are paths; its boundary data contain the endpoint state values, which are complete in the elementary Markov case; and gluing integrates over the intermediate state. A finite lattice block or a spacetime subdomain in a field theory can also be a region once the exposed boundary data and gluing measure have been specified.

The regular smooth assumptions used below play the role that familiar regularity assumptions play in ordinary mechanics: finite-dimensional smooth spaces, differentiable actions or generators, and nondegeneracy where one wants a local map rather than a relation. The direction, however, is different. Ordinary Lagrangian, Hamiltonian, and Hamilton-Jacobi mechanics usually start with an action, Hamiltonian, or symplectic phase space; here exact gluing of weights is assumed first, and the smooth generator and response geometry are reconstructed from it in the sharp limit.

If R is decomposed into subregions along an internal boundary Σ ,

$$R = R_1 \cup_{\Sigma} R_2,$$

then the cut-objective requirement is the exact gluing law

$$K_R(\varphi_{\partial R}) = \int_{\mathcal{B}_{\Sigma}} K_{R_1}(\varphi_{\partial R_1}) K_{R_2}(\varphi_{\partial R_2}) \mu_{\Sigma}(d\varphi_{\Sigma}). \quad (1)$$

Here $\mathcal{B}_{\Sigma}$ is the space of admissible data on the internal cut, and μ_{Σ} is the measure used to sum over those data. The boundary data on ∂R_1 and ∂R_2 agree with the external data on ∂R and with the common internal value φ_{Σ} . This gluing law is a structural assumption on the family of region weights used for the reconstruction. In concrete theories it may be verified or derived from Chapman-Kolmogorov equations, path-integral composition, transfer matrices, topological field theory, and general-boundary quantum theory [25, 26, 4, 5]. The gluing formula alone does not supply the reconstruction. Once such weights have a regular sharp generator, differentiating that generator supplies the cotangent response geometry normally postulated as phase space.

This distinction between taking W as a primitive endpoint generator and deriving W from composable weights is central. Starting from a differentiable function $W(a, b)$ recovers ordinary generating-function theory. Starting instead from boundary-conditioned weights K_R , exact gluing, and a sharp limit makes W the leading generator of composable predictive weights. The construction is a conditional bridge from composable history weights to the boundary-response variables used in Hamilton-Jacobi and generating-function mechanics.

The construction separates three layers which are often conflated.

1. Exact composition of weights: a whole-region weight is obtained by summing or integrating over complete internal boundary data.
2. Cut-objectivity: an auxiliary cut is a representation of a history fragment, not an additional physical event, so predictions for the whole region must not depend on its admissible placement.
3. Sharp-limit geometry: only when a differentiable large-deviation or stationary-phase generator exists do boundary-response covectors and Lagrangian relations appear.

The resulting claim is:

regular sharp cut-objective predictive weights $\Rightarrow$ boundary-response phase geometry.

Without exact gluing, completeness, a sharp generator, smoothness, and nondegeneracy or a relation-level formulation, the advertised boundary-response geometry does not follow; in particular, no symplectic phase space follows.

The remaining sections are organized as follows. Section 2 makes the preceding compositional language precise by defining regions, auxiliary cuts, complete boundary proposals, predictive boundary equivalence, and cut-objective composition. Section 3 derives sharp-generator composition from positive gluing through a Laplace principle and records the local oscillatory analogue. Section 4 defines boundary response, proves response matching at an internal cut, and shows that a twice differentiable endpoint generator defines a Lagrangian response relation, with a local symplectic map in the mixed-Hessian nondegenerate case. Section 5 gives Brownian and Ornstein-Uhlenbeck transition-kernel examples. Section 6 discusses the conditional claim, its relation to existing frameworks, and its limitations. Section 7 concludes.

2 regular compositional setup

2.1 regions, cuts, and boundary data

We prove the main theorem in the following finite-dimensional setting. The theorem does not require a universal notion of spacetime domain. It requires composable units with interior histories, boundary data, region weights, and an associative gluing operation. Field-theoretic and quantum cases require additional structure beyond this regular finite-dimensional setting.

Definition 1 (admissible decomposition). *Relative to a chosen class of regions, weights, observations, and allowed cuts, a decomposition $R = R_1 \cup_{\Sigma} R_2$ is admissible when R_1 and R_2 are allowed regions, Σ is an allowed internal interface, the boundary data on R , R_1 , R_2 , and Σ restrict consistently, and an interface-data space $\mathcal{B}_{\Sigma}$ with gluing measure μ_{Σ} has been specified so that the gluing integral is well defined. Not every formal geometric split of a domain is assumed to be admissible.*

Assumption 1 (regular compositional setting). *There is a class $\mathcal{R}$ of bounded composable regions. Each region $R \in \mathcal{R}$ has an interior history space $\mathcal{H}_R$, a boundary ∂R , and a boundary-data space $\mathcal{B}_{\partial R}$. For each admissible decomposition $R = R_1 \cup_{\Sigma} R_2$, the shared internal interface is Σ , its interface-data space is $\mathcal{B}_{\Sigma}$, and there are restriction maps from the external and internal boundary data to the boundary-data spaces of R_1 and R_2 . There is also a measure μ_{Σ} on $\mathcal{B}_{\Sigma}$ used to sum or integrate over the data exposed on the shared interface. The gluing operation is associative whenever three or more regions are composed and the relevant integrals are defined.*

In this setting a region weight is a function

$$K_R : \mathcal{B}_{\partial R} \rightarrow [0, \infty)$$

or a density relative to a specified reference measure. When such a representation is available, a continuum model may write this boundary weight formally as an integral over compatible interior histories:

$$K_R(\varphi_{\partial R}) = \int_{\{\gamma \in \mathcal{H}_R : \gamma|_{\partial R} = \varphi_{\partial R}\}} w_R[\gamma] \mathcal{D}\gamma.$$

Here γ is an interior history, $w_R[\gamma]$ is its model-dependent weight, and $\mathcal{D}\gamma$ denotes the corresponding formal measure over histories. This formula is only a representation of K_R , and the theorem does

not assume that every K_R necessarily has such a representation. The theorem assumes that the boundary weight K_R and its gluing law are well defined; it does not require a unique or universal construction of $w_R[\gamma] \mathcal{D}\gamma$. Different microscopic representations that produce the same K_R are equivalent for the reconstruction.

Definition 2 (auxiliary cut). *For an admissible decomposition $R = R_1 \cup_\Sigma R_2$, the internal boundary Σ is auxiliary when it is introduced only to compute or represent the region weight. It is not a preparation device, detector, material wall, permanent record, or additional external condition.*

Definition 3 (complete boundary proposal). *A boundary proposal on an internal interface Σ consists of a proposed boundary-description space $\mathcal{C}_\Sigma$ and a variable $\chi_\Sigma \in \mathcal{C}_\Sigma$. The space $\mathcal{C}_\Sigma$ is not assumed to be the admissible interface-data space $\mathcal{B}_\Sigma$ from the regular setup; it is a candidate description whose adequacy is being tested. To use such a proposal in a gluing formula, one must specify how it supplies admissible subregion boundary data. Thus, for each admissible decomposition in the class, assume there are measurable comparison maps*

$$\iota_i : \mathcal{B}_{\partial R} \times \mathcal{C}_\Sigma \rightarrow \mathcal{B}_{\partial R_i}, \quad i = 1, 2,$$

which combine the external boundary data with the proposed internal description. In the ordinary case $\mathcal{C}_\Sigma = \mathcal{B}_\Sigma$, the maps ι_i are just the restriction maps from external data and the shared interface value to the two subregion boundaries. The proposal is complete for a class of regions and observations if there is a measure ν_Σ such that

$$K_R(\varphi_{\partial R}) = \int_{\mathcal{C}_\Sigma} K_{R_1}(\iota_1(\varphi_{\partial R}, \chi_\Sigma)) K_{R_2}(\iota_2(\varphi_{\partial R}, \chi_\Sigma)) \nu_\Sigma(d\chi_\Sigma)$$

for every admissible decomposition in the class. A complete proposal is minimal relative to that class if no nontrivial measurable or smooth projection of $\mathcal{C}_\Sigma$ still gives the same gluing law for every decomposition, except for discarding null variables or making an invertible relabelling.

Here a null variable is one that has no effect on the region weights, up to sets ignored by the gluing measure. This qualification prevents artificial failures of minimality: adding a dummy coordinate to χ_Σ , or changing coordinates on $\mathcal{C}_\Sigma$, should not count as adding genuine boundary content.

Definition 4 (predictive boundary equivalence). *Fix a class of admissible regions, cuts, external observations, and region weights. Two boundary descriptions $\chi_\Sigma \in \mathcal{C}_\Sigma$ and $\chi'_\Sigma \in \mathcal{C}'_\Sigma$ are predictively equivalent when replacing one by the other in every admissible gluing construction gives the same whole-region weights for all external boundary data in the class. A predictive boundary state, in this setting, is an equivalence class of boundary descriptions under this relation.*

The replacement in this definition is always mediated by the comparison maps that feed each candidate boundary description into the same subregion boundary-data spaces. Predictive equivalence is therefore not an intrinsic property of $\mathcal{C}_\Sigma$ alone. It is relative to the chosen admissible regions, cuts, weights, comparison maps, and observation class. Enlarging the class of external observations or admissible decompositions can refine the equivalence relation, while restricting that class can coarsen it.

The two preceding definitions separate three notions. First, one proposes boundary variables. Second, one verifies whether those variables make the gluing formula true for the admissible decompositions under consideration. Third, one asks whether any proposed variables are redundant. Completeness is therefore not simply the statement that gluing works by definition. It is a property of a boundary proposal relative to a specified class of admissible decompositions, region weights, and

observations. The formal sufficiency notion used here is gluing sufficiency for boundary-conditioned region weights. It is therefore different from the causal state of computational mechanics and from a Bayesian filtering state: no separate theorem about optimal forecasting, filtering, or statistical sufficiency is assumed or proved here.

Principle 1 (cut-objectivity). *For any physical process represented by a region R , and for any admissible auxiliary decomposition $R = R_1 \cup_{\Sigma} R_2$, the physical weight assigned to the whole region is independent of the insertion, displacement, refinement, or removal of the internal cut, provided the variables summed or integrated over on Σ form a complete boundary proposal for the specified class of admissible decompositions, weights, and observations.*

Cut-objectivity uses the property of completeness as a condition. In this paper it is a modelling postulate about admissible auxiliary decompositions: when the interface proposal is complete, inserting that auxiliary interface does not change the whole-region boundary weight. The analogy with coordinate or gauge independence is only an analogy. A coordinate chart may help describe a system without changing invariant quantities; similarly, an auxiliary cut may help factor a region without changing the whole-region weight. The principle does not apply to a real intervention, measurement, material boundary, or changed external condition; those define a different physical process or a different region weight. In the formal results below, the operational content of cut-objectivity is the named exact positive gluing assumption. The paper does not prove invariance under arbitrary geometric displacements or refinements of cuts; admissibility and exact gluing are part of the chosen class of regions, cuts, weights, and observations.

2.2 operational tests for completeness

The practical test is direct:

$$K_R(\varphi_{\partial R}) \neq \int K_{R_1} K_{R_2} d\mu_{\Sigma} \\ \Rightarrow \text{the proposed boundary description is incomplete, inadmissible, or wrong.}$$

This test is useful because failures have recognizable forms. The boundary description is incomplete when the interface data are missing variables needed for gluing. The cut is inadmissible when the proposed decomposition is not allowed for the theory or observational class. The modelling setup is wrong when the chosen region weight, gluing measure, or class of histories is not the object that actually composes.

The following examples illustrate the role of boundary data. In each case the question is the same: what must be exposed on the interface so that the two sides can be recombined without changing the whole-region weight?

- For a Markov diffusion X_t , the current state X_t is complete for future transition probabilities: conditioning on the full current state makes earlier history irrelevant for future kernels. If only a coarse observed coordinate $Y_t = h(X_t)$ is retained, that coordinate need not be Markov. Then Y_t alone may fail the gluing test, and one must add memory variables or hidden coordinates to recover a composable description.
- A finite-memory chain gives the same test in a discrete toy model. If X_n is second-order Markov, with transition probabilities $P(x_{n+1} | x_n, x_{n-1})$, then retaining only x_n at a cut does not generally define a right-hand factor independent of the left history. For example, the two-step weight contains

$$P(x_1 | x_0)P(x_2 | x_1, x_0),$$

so the second factor still depends on x_0 . Unless $P(x_2 | x_1, x_0)$ is independent of x_0 , gluing over x_1 alone fails. A complete Markovian boundary description for arbitrary cuts must carry the memory variable, for instance the pair (x_{n-1}, x_n) . This illustrates the sense in which the cut-objective test can force extra boundary content.

- In a field theory, the required interface data depend on the action, constraints, and boundary conditions. Field values may be enough in one formulation, while another may require normal derivatives, fluxes, constraint data, or boundary-condition data to compose regions correctly. This is the same lesson expressed in variational and covariant phase-space treatments: boundary terms and boundary conditions determine what data are actually exposed at an interface [22, 17].
- In gauge theory, cutting a subsystem can expose gauge degrees of freedom at the boundary. To compose local subsystems without losing gauge invariance, one may need edge modes or gauge frames as part of the interface data [11].
- In an open stochastic system, coarse boundary data may hide entropy production or time-asymmetric information. Such coarse variables need not compose as closed Markov data, and the loss of a closed description may have to be represented by additional boundary variables, memory terms, or an explicitly open-system account [12].

Thus a complete predictive boundary is not always just a configuration value. Here a configuration value means the kinematic variable directly fixed on the interface, such as an endpoint position q or a field value on a boundary. A complete predictive boundary is broader: it is whatever variable set makes the gluing law true for the chosen observational class. In regular endpoint examples, exact gluing may use configuration data q on the interface, while the sharp limit attaches a response covector p by differentiating the resulting generator. The pair (q, p) is then a reconstructed phase-response variable, not automatically the data integrated over in the original finite- ε gluing formula. In singular, non-Markovian, or gauge systems the same gluing test may instead lead to constrained, memory-augmented, flux-augmented, or presymplectic boundary data.

3 sharp generators

3.1 positive large-deviation weights

Assumption 2 (exact positive gluing). *For each admissible decomposition $R = R_1 \cup_\Sigma R_2$ in the class under consideration and each $\varepsilon > 0$, the region weights are positive on the boundary-data domain being studied, or else are interpreted with the extended rate convention $-\varepsilon \log 0 = +\infty$. The weights satisfy the exact gluing law*

$$K_R^\varepsilon(\varphi_{\partial R}) = \int_{\mathcal{B}_\Sigma} K_{R_1}^\varepsilon(\varphi_{\partial R_1}) K_{R_2}^\varepsilon(\varphi_{\partial R_2}) \mu_\Sigma^\varepsilon(d\varphi_\Sigma).$$

In this formula, $\mathcal{B}_\Sigma$ is the space of admissible data on the internal cut, and μ_Σ^ε is the measure over $\mathcal{B}_\Sigma$ used in the gluing integral. The subregion boundary data $\varphi_{\partial R_1}$ and $\varphi_{\partial R_2}$ are determined by the external boundary data, the internal interface value, and the restriction maps from the regular compositional setting.

This assumption is the formal input corresponding to cut-objective composition in the positive setting. The following Laplace principle is the substantive analytic hypothesis that turns exact finite- ε gluing into an infimum rule for sharp generators.

The parameter ε is a sharpness parameter. In examples it may be a noise strength, a temperature scale, an inverse system-size scale, or a semiclassical parameter; the formal role is that smaller ε makes the weight more strongly concentrated on the dominant histories. The large-deviation form is not an exact equality at finite ε . It means the logarithmic asymptotic statement

$$K_R^\varepsilon(\varphi_{\partial R}) = \exp\left[-\frac{W_R(\varphi_{\partial R}) + o(1)}{\varepsilon}\right] \quad (\varepsilon \rightarrow 0),$$

or equivalently

$$-\varepsilon \log K_R^\varepsilon(\varphi_{\partial R}) = W_R(\varphi_{\partial R}) + o(1) \quad (\varepsilon \rightarrow 0).$$

This formulation ignores only subexponential factors, since such factors disappear after applying $-\varepsilon \log$. We assume the sharp generator exists:

$$W_R(\varphi_{\partial R}) = -\lim_{\varepsilon \rightarrow 0} \varepsilon \log K_R^\varepsilon(\varphi_{\partial R}). \quad (2)$$

Large-deviation theory supplies many settings in which probabilities or density kernels acquire exponential rate functions in small-noise, long-time, many-particle, or thermodynamic limits [16, 13, 29]. Maximum caliber and path-entropy methods provide related variational constructions of path weights [27, 10]. Schrödinger bridge theory formulates endpoint-constrained stochastic evolution through relative-entropy minimisation over path laws [9, 21].

Assumption 3 (Laplace principle with external uniformity). *For the admissible decomposition $R = R_1 \cup_\Sigma R_2$, let $x = \varphi_{\partial R}$ denote external boundary data and let $y = \varphi_\Sigma \in \mathcal{B}_\Sigma$ denote internal interface data. Assume:*

1. *the internal boundary-data space used in the sharp limit is a locally compact finite-dimensional smooth manifold, or a compact subset of one;*
2. *for each x in a neighbourhood V of the external data under consideration, the subregion restrictions determine arguments (x_1, y) and (y, x_2) for R_1 and R_2 ;*

3. *the functions*

$$f_\varepsilon(x, y) = -\varepsilon \log[K_{R_1}^\varepsilon(x_1, y)K_{R_2}^\varepsilon(y, x_2)]$$

converge on the relevant sublevel sets to $f(x, y) = W_{R_1}(x_1, y) + W_{R_2}(y, x_2)$;

4. *the gluing measures μ_Σ^ε have an exponential rate $I_\Sigma : \mathcal{B}_\Sigma \rightarrow [0, \infty]$. Informally, near an interface value y , this means*

$$\mu_\Sigma^\varepsilon(dy) \sim \exp[-I_\Sigma(y)/\varepsilon] \times (\text{subexponential factor}).$$

For example, relative to a reference measure $\lambda(dy)$, a density of the form

$$\mu_\Sigma^\varepsilon(dy) = \exp[-I_\Sigma(y)/\varepsilon] \rho_\varepsilon(y) \lambda(dy), \quad -\varepsilon \log \rho_\varepsilon(y) \rightarrow 0,$$

has rate I_Σ . The precise operative assumption is the logarithmic integral asymptotic below. When the measures are subexponential at scale ε , $I_\Sigma = 0$;

5. *the limiting cost $f(x, y) + I_\Sigma(y)$ is lower semicontinuous in y and has compact sublevel sets on the internal variables relevant to $x \in V$;*

6. the logarithmic integral asymptotic

$$-\varepsilon \log \int_{\mathcal{B}_\Sigma} \exp[-f_\varepsilon(x, y)/\varepsilon] \mu_\Sigma^\varepsilon(dy) \longrightarrow \inf_{y \in \mathcal{B}_\Sigma} [f(x, y) + I_\Sigma(y)]$$

holds locally uniformly for $x \in V$.

Locally compact means that each point has a compact neighbourhood; allowing a compact subset of a smooth manifold covers common constrained domains and boundary-data ranges. The final item is the precise Laplace principle applied to the gluing integral over $\mathcal{B}_\Sigma$. It combines the sharp limits of the two subregion weights with any exponential rate carried by the gluing measure, and it rules out escape of the dominant contribution in the internal variable.

Lemma 1 (sharp composition). *Under exact positive gluing in Assumption 2, the sharp-generator limit (2), and the Laplace principle above,*

$$W_R(\varphi_{\partial R}) = \inf_{\varphi_\Sigma \in \mathcal{B}_\Sigma} [W_{R_1}(\varphi_{\partial R_1}) + W_{R_2}(\varphi_{\partial R_2}) + I_\Sigma(\varphi_\Sigma)]. \quad (3)$$

If μ_Σ^ε is subexponential at scale ε , then $I_\Sigma = 0$ and this reduces to the two-cost formula.

Proof. For fixed external boundary data, exact gluing writes K_R^ε as an integral over internal boundary data. The stated Laplace principle identifies the exponential rate of that integral with the infimum of the sum of the two subregion rates plus the measure rate I_Σ . Taking $-\varepsilon \log$ and passing to the limit gives (3). Local uniformity ensures that the resulting function is the sharp generator for nearby external data, not merely a pointwise lower-semicontinuous envelope. $\square$

For positive exponential weights, sharp composition gives an infimum of the combined cost. Response matching becomes ordinary derivative equality only when that infimum is attained at a smooth interior minimizer. Boundary minimizers, corners, or several coexisting minimizers require the corresponding relation-level formulation, using subgradients or normal cones where appropriate.

3.2 oscillatory weights

The main theorem uses positive weights. The oscillatory case below is only a local stationary-phase analogue, not an extension of the theorem to quantum amplitudes. In an oscillatory regime one may have

$$K_R(\varphi_{\partial R}) \sim A_R(\varphi_{\partial R}) \exp \left[\frac{i}{\hbar} S_R(\varphi_{\partial R}) \right].$$

The corresponding gluing formula is then a coherent sum of amplitudes over intermediate data, so phases can interfere. Stationary phase gives a single phase composition law only under additional local assumptions: a chosen branch, isolated nondegenerate critical point, no caustic in the patch considered (that is, no local merger or singular focusing of stationary points), and controlled Maslov phases and prefactor contributions. Together, these assumptions say that the gluing integral has one smooth local stationary contribution at leading phase order. Away from stationary internal data, the rapidly oscillating phases cancel at leading order. Here $\text{stat}_{\varphi_\Sigma}$ means evaluation at stationary values with respect to the intermediate data φ_Σ ; it is not a minimum. At leading phase order, under those restrictions one may write

$$S_R(\varphi_{\partial R}) = \text{stat}_{\varphi_\Sigma} [S_{R_1}(\varphi_{\partial R_1}) + S_{R_2}(\varphi_{\partial R_2})].$$

Once such a single smooth branch exists, the derivative argument used in Section 4 applies to S_R with stationarity replacing minimization. Without the branch restriction, however, several critical points may interfere and no single smooth phase generator need exist. The formal theorem is therefore stated for positive sharp generators: positivity gives a genuine variational composition by an infimum, while oscillatory amplitudes give only this local stationary-phase analogue related to path integrals and the general-boundary formulation [14, 26].

4 boundary response and response geometry

4.1 boundary response from sharp generators

This section starts from smooth generators that already satisfy a variational composition law. In the main line of the paper, those generators are supplied by the positive gluing and Laplace-principle assumptions of Section 3. The same local calculus can also describe a generator that comes from ordinary mechanics, optimal control, a bridge problem, or a single smooth oscillatory branch.

Let $\mathcal{M}$ be a smooth finite-dimensional boundary-configuration manifold, and let $a, b, c \in \mathcal{M}$ denote boundary configurations on three consecutive cuts. Let $W_1(a, b)$ and $W_2(b, c)$ be sharp generators for the two adjacent regions. The sharp-composition lemma gives the more general local form

$$W(a, c) = \inf_b [W_1(a, b) + W_2(b, c) + I(b)], \quad (4)$$

where I is the interface measure rate. In the common subexponential case $I = 0$, this reduces to the two-cost composition formula.

Definition 5 (boundary response). *The boundary response of a C^1 sharp generator with respect to a boundary configuration $q \in \mathcal{M}$ is the covector obtained by differentiating the generator with respect to q , with the sign determined by whether the boundary is incoming or outgoing for the oriented region.*

For the two-segment composition define

$$p_b^- := D_b W_1(a, b), \quad p_b^+ := -D_b W_2(b, c).$$

The minus sign identifies the same internal boundary with opposite orientations as seen from the two adjacent regions.

The first-order condition for an interior minimizer therefore gives a response balance law:

Proposition 1 (internal response balance). *Assume W_1 , W_2 , and I are C^1 in the internal variable near b^* , and that the infimum in (4) is attained at a point b^* in the interior of the relevant finite-dimensional manifold or chart domain. Then the oriented boundary responses, evaluated at the common internal cut, satisfy*

$$p_b^-(a, b^*) - p_b^+(b^*, c) = -DI(b^*).$$

In particular, if the gluing measure is subexponential at the sharp scale so that $I = 0$, or if $DI(b^) = 0$, the oriented responses match:*

$$p_b^-(a, b^*) = p_b^+(b^*, c).$$

Proof. Since b^* is an interior minimizer and the objective is differentiable near b^* , Fermat's rule gives

$$D_b W_1(a, b^*) + D_b W_2(b^*, c) + DI(b^*) = 0.$$

Using $p_b^- = D_b W_1(a, b^*)$ and $p_b^+ = -D_b W_2(b^*, c)$, this is

$$p_b^- - p_b^+ = -DI(b^*).$$

The stated matching condition follows when the rate term has zero derivative at b^* , in particular when $I = 0$. $\square$

This is the regular interior case. If the minimizer is constrained or nonsmooth, the corresponding statement involves subdifferentials or normal cones rather than equality of ordinary derivatives.

For a single history fragment with two boundary configurations $a, b \in \mathcal{M}$, let $W(a, b)$ be a C^2 sharp generator on an open set of $\mathcal{M} \times \mathcal{M}$. Define endpoint responses by

$$p_a = -D_a W(a, b), \quad p_b = D_b W(a, b). \quad (5)$$

Then

$$dW(a, b) = -p_a \cdot da + p_b \cdot db. \quad (6)$$

The pair (q, p) is not postulated as a mechanical phase point. It is the boundary configuration together with the first-order response of the sharp predictive generator to a boundary displacement.

4.2 regular response geometry

In local coordinates $(q^1, \dots, q^n)$ on $\mathcal{M}$, with covector components $(p_1, \dots, p_n)$, let

$$\theta = \sum_{i=1}^n p_i dq^i$$

be the canonical one-form on the cotangent bundle $T^*\mathcal{M}$. Use the canonical symplectic form [3, 1]

$$\omega = -d\theta = \sum_{i=1}^n dq^i \wedge dp_i.$$

Write $T^*\mathcal{M}^-$ for $T^*\mathcal{M}$ equipped with $-\omega$. Here “Lagrangian” is used in the standard symplectic-geometric sense: the product symplectic form vanishes on the relation, and the relation has half the dimension of the ambient product phase space [3, 1].

Theorem 1 (endpoint response relation is Lagrangian). *Let $\mathcal{M}$ be a smooth n -dimensional boundary-configuration manifold and let $W \in C^2(U)$ for an open set $U \subset \mathcal{M} \times \mathcal{M}$. Define*

$$\Phi_W : (a, b) \mapsto (a, -D_a W(a, b), b, D_b W(a, b)) \quad (7)$$

and the endpoint response relation

$$L_W = \{(a, p_a, b, p_b) : p_a = -D_a W(a, b), p_b = D_b W(a, b)\} \subset T^*\mathcal{M}^- \times T^*\mathcal{M}$$

Then $L_W = \Phi_W(U)$ is an embedded Lagrangian submanifold.

Proof. Because Φ_W includes the base point (a, b) , its image is the graph over U of the C^1 covector-valued map $(a, b) \mapsto (-D_a W(a, b), D_b W(a, b))$. Thus L_W is an embedded $2n$ -dimensional submanifold of $T^*\mathcal{M}^- \times T^*\mathcal{M}$. Pull back the one-form $\theta_a - \theta_b$ on $T^*\mathcal{M}^- \times T^*\mathcal{M}$ to U along the endpoint response map (7). By (6), the pullback is $-dW$. Therefore the pullback of the product symplectic form $-\omega_a + \omega_b = d\theta_a - d\theta_b$ is

$$d(-dW) = 0.$$

The image is isotropic. Since U has dimension $2n$, which is half the dimension of $T^*\mathcal{M}^- \times T^*\mathcal{M}$, the embedded relation is Lagrangian. $\square$

Corollary 1 (symplectic-map case). *If, in addition, the derivative $\partial_b(-D_a W) : T_b \mathcal{M} \rightarrow T_a^* \mathcal{M}$, the mixed Hessian block in local coordinates, is invertible at the point on the branch considered, then there are open neighbourhoods $O_-, O_+ \subset T^* \mathcal{M}$ and the endpoint response relation L_W locally defines a map*

$$F : O_- \rightarrow O_+, \quad (a, p_a) \mapsto (b, p_b),$$

and F preserves the canonical symplectic form:

$$F^* \omega = \omega.$$

Proof. Invertibility of $\partial_b(-D_a W)$ implies, by the implicit-function theorem, that the equation $p_a = -D_a W(a, b)$ locally determines b as a function of (a, p_a) . Substitution into $p_b = D_b W(a, b)$ gives the local map F . The graph of F is the corresponding branch of the Lagrangian relation L_W , so $-\omega_a + \omega_b$ vanishes on the graph. Equivalently, $F^* \omega = \omega$ on O_- . $\square$

Theorem 2 (regular sharp compositional reconstruction). *Consider a regular finite-dimensional endpoint setting in which the exact positive gluing assumption, the sharp-generator limits, and the locally uniform Laplace principle hold for the admissible cuts under consideration. Suppose that the interface measure is subexponential at the sharp scale for the response-matching step, so that $I = 0$. Suppose also that each internal cut where response matching is asserted satisfies the hypotheses of Proposition 1: the relevant variational objective is C^1 in the internal variable near an attained smooth interior minimizer b^* . If the resulting endpoint generator W is C^2 on an open set $U \subset \mathcal{M} \times \mathcal{M}$, then the boundary derivatives*

$$p_a = -D_a W(a, b), \quad p_b = D_b W(a, b)$$

define an embedded Lagrangian endpoint response relation

$$L_W \subset T^* \mathcal{M}^- \times T^* \mathcal{M}.$$

If the mixed Hessian block $\partial_b(-D_a W)$ is invertible at a point, the relation is locally the graph of a symplectic map between open subsets of $T^ \mathcal{M}$. At any internal cut satisfying the stated smooth interior-minimizer hypotheses, the oriented responses match when $I = 0$.*

If the interface measure has a nonzero differentiable rate I and the same smooth interior-minimizer hypotheses hold, the interior-cut conclusion is the response balance

$$p_b^- - p_b^+ = -DI(b^*)$$

rather than unmodified response equality, unless the rate is absorbed into the chosen subregion generators by an explicit convention.

Proof. Assumption 2 and the Laplace principle give the sharp-composition lemma, hence variational composition of the sharp generator. When $I = 0$ and the stated C^1 smooth interior-minimizer hypotheses hold, the response-balance proposition gives equality of oriented responses at the corresponding cuts. The endpoint response relation is Lagrangian by the preceding theorem. The local symplectic-map statement follows from the mixed-Hessian corollary. $\square$

The exactness of dW is the reason the canonical two-form is preserved. This is the standard generating-function argument in symplectic geometry and Hamilton-Jacobi theory [3, 24]. Here the same argument is applied after W has been obtained as the sharp generator of cut-objective predictive weights, so the cotangent variables are reconstructed as boundary responses rather than inserted as primitive mechanical momenta.

4.3 presymplectic and constrained cases

The cotangent bundle $T^*\mathcal{M}$ is the clean regular model. If the endpoint response relation is multivalued, singular, gauge-redundant, or constrained, the natural output is a Lagrangian relation, a presymplectic manifold, or a reduced constrained phase space rather than a global symplectic map. This is expected in singular Lagrangian systems, gauge theories, and field theories with boundaries. Covariant phase-space methods show that boundary terms, corner choices, and boundary conditions can affect the boundary symplectic structure [22, 17].

For the main reconstruction path of this paper, the hierarchy should therefore be read as

exact composable weights + complete auxiliary cuts
+ sharp-limit regularity
→ sharp generator
→ boundary response relation
→ Lagrangian or presymplectic structure
→ symplectic phase space in regular nondegenerate cases.

5 worked examples

5.1 brownian transition kernel

The Brownian example begins with a probabilistic transition kernel, not a mechanical action. For notational simplicity this section uses the one-dimensional kernel; the d -dimensional version used for orientation in the introduction is the corresponding Gaussian heat kernel with determinant normalisation and the same componentwise response calculation. Let

$$K_\varepsilon(a, b; T) = \frac{1}{\sqrt{2\pi\varepsilon T}} \exp\left[-\frac{(b-a)^2}{2\varepsilon T}\right], \quad T > 0.$$

This is the heat kernel for one-dimensional Brownian motion with noise strength ε , and it composes exactly by the Chapman-Kolmogorov relation [20]:

$$K_\varepsilon(a, c; T_1 + T_2) = \int_{\mathbb{R}} K_\varepsilon(a, b; T_1) K_\varepsilon(b, c; T_2) db. \quad (8)$$

The sharp generator is

$$W_T(a, b) = \frac{(b-a)^2}{2T}.$$

The variational composition is

$$W_{T_1+T_2}(a, c) = \inf_b \left[\frac{(b-a)^2}{2T_1} + \frac{(c-b)^2}{2T_2} \right],$$

with minimizer

$$b^* = \frac{T_2 a + T_1 c}{T_1 + T_2}.$$

The oriented internal responses are

$$p_b^- = \frac{b^* - a}{T_1}, \quad p_b^+ = \frac{c - b^*}{T_2},$$

and the minimizer gives $p_b^- = p_b^+$.

For a single interval,

$$p_a = -D_a W_T(a, b) = \frac{b - a}{T}, \quad p_b = D_b W_T(a, b) = \frac{b - a}{T}.$$

Thus $p_a = p_b = p$ and

$$b = a + Tp.$$

The induced response map is

$$F_T(a, p) = (a + Tp, p).$$

Using the convention $\omega = dq \wedge dp$, the coordinate check is

$$F_T^*(db \wedge dp_b) = d(a + Tp) \wedge dp = da \wedge dp,$$

so the canonical two-form is preserved. The usual free-particle symplectic map has appeared from the sharp limit of a composable stochastic kernel. In this Markov example the predictive boundary state for gluing is represented by the endpoint value itself, up to predictive equivalence of boundary descriptions. The response covector is not integrated over in the finite ε kernel; it is reconstructed later as the gradient of the sharp generator.

5.2 ornstein-uhlenbeck transition kernel

A slightly less trivial positive example is the small-noise Ornstein-Uhlenbeck diffusion

$$dX_t = \alpha X_t dt + \sqrt{\varepsilon} dB_t.$$

Its transition density is Gaussian with mean $e^{\alpha T} a$ and variance εq_T , where

$$q_T = \begin{cases} \frac{e^{2\alpha T} - 1}{2\alpha}, & \alpha \neq 0, \\ T, & \alpha = 0. \end{cases}$$

Thus

$$K_\varepsilon(a, b; T) = \frac{1}{\sqrt{2\pi\varepsilon q_T}} \exp \left[-\frac{(b - e^{\alpha T} a)^2}{2\varepsilon q_T} \right],$$

and the sharp generator is

$$W_T(a, b) = \frac{(b - e^{\alpha T} a)^2}{2q_T}.$$

The Gaussian Markov property gives exact composition for all $T_1, T_2 > 0$, so the sharp generators compose by the infimum principle.

The endpoint responses are

$$p_b = D_b W_T(a, b) = \frac{b - e^{\alpha T} a}{q_T}, \quad p_a = -D_a W_T(a, b) = e^{\alpha T} p_b.$$

Consequently the response map is

$$p_b = e^{-\alpha T} p_a, \quad b = e^{\alpha T} a + q_T e^{-\alpha T} p_a.$$

This map is not the free Brownian shear. It expands the boundary configuration by $e^{\alpha T}$ and contracts the response by $e^{-\alpha T}$, with an additional shear determined by q_T . Using the same convention $\omega = dq \wedge dp$, directly,

$$db \wedge dp_b = (e^{\alpha T} da + q_T e^{-\alpha T} dp_a) \wedge e^{-\alpha T} dp_a = da \wedge dp_a.$$

The canonical response geometry is again preserved. The example shows that the construction is not limited to free Brownian motion: a positive stochastic kernel with drift produces a nontrivial canonical relation in its sharp boundary-response variables. As in the Brownian case, the complete predictive boundary state for Markov gluing is the endpoint value, while the response covector is a sharp-limit derivative attached after the logarithmic generator has been formed.

6 discussion

6.1 claim and scope

The main claim is conditional:

$$\begin{array}{lcl} \text{exact composable weights} & + & \text{complete auxiliary cuts} \\ + \text{ regular sharp generator} & \Rightarrow & \text{boundary-response geometry.} \end{array}$$

At the relation level, the output is Lagrangian. Under mixed-Hessian nondegeneracy, the output is a local symplectic map. Cut-objectivity supplies the compositional input through exact gluing; the sharp generator and regularity hypotheses supply the response geometry.

The conceptual payoff is a controlled reinterpretation of phase variables. The configuration is boundary data retained for gluing. The conjugate covector is the first-order response of the sharp generator to boundary displacement. When a differentiable one-parameter family of endpoint generators satisfies a composition law and the usual regular nondegeneracy hypotheses, the resulting local endpoint maps are canonical; deriving a continuous Hamiltonian flow from such a family requires those additional hypotheses and is not part of the finite-region theorem proved here.

This locates the paper among familiar constructions. The gluing equation appears in Markov kernels, propagators, transfer matrices, topological field theory, path integrals, and process theories. The symplectic step is ordinary generating-function theory. The contribution is the route between them: composable weights give a sharp generator, and that generator gives boundary responses. In positive large-deviation examples those responses may be costates, thermodynamic forces, or optimization multipliers rather than physical momenta. The interpretation as canonical momentum requires a mechanical context in which the response relation has that meaning. Quantum mechanics, spacetime, and gravity require additional structure beyond the regular positive sharp setting studied here.

6.2 relation to existing frameworks

The nearest comparisons play different roles. Predictive-state and statistical sufficiency formalisms clarify what is and is not meant by prediction here; large-deviation, maximum-caliber, bridge, and control theories supply nearby sources of sharp generators; Hamilton-Jacobi and variational mechanics supply the endpoint-response geometry; and general-boundary, path-integral, open Markov, and gauge-theoretic settings clarify the compositional and boundary aspects of the construction.

Computational mechanics defines causal states as equivalence classes of pasts with identical conditional distributions over futures, and proves minimality results for prediction [28]. Predictive

State Representations similarly treat state as predictive content in controlled stochastic systems [23]. Bayesian filtering supplies a different predictive-state paradigm: a conditional distribution or finite filtering statistic is propagated as observations arrive [19, 7]. Information geometry studies statistical-model manifolds and their natural geometric structures [2]. Standard statistical sufficiency concerns data reductions that preserve the relevant statistical model or likelihood [8]. The sufficiency target here is different: a complete boundary is sufficient for gluing history weights across admissible cuts, not merely for forecasting, filtering, or likelihood-based inference.

Large-deviation theory explains how probability weights generate rate functions and Hamilton-Jacobi structures in sharp limits [16, 13, 29]. Maximum caliber and path-entropy principles provide variational constructions of path distributions [27, 10]. Schrödinger bridge theory supplies a related case in which endpoint constraints and path-space entropy costs generate dual potentials [18, 9, 21]. Stochastic control and dynamic programming similarly produce value functions governed by Hamilton-Jacobi-Bellman equations [15]. These literatures supply possible origins for a sharp generator W . The additional step here is to interpret endpoint derivatives of such a generator as boundary responses associated with cut-objective composition.

Hamilton-Jacobi theory already identifies momenta with derivatives of Hamilton’s principal function [3]. Discrete variational mechanics shows that action sums over adjacent time steps generate symplectic maps [24]. Those facts supply the local symplectic mechanism. The distinction is the starting point: W is not postulated as a mechanical action but obtained as the sharp generator of composable predictive weights.

The general-boundary formulation assigns state spaces to boundaries and amplitudes to spacetime regions [25, 26]. Path integrals compose by summing over intermediate boundary configurations when an identity resolution is inserted [14]. These are close relatives of cut-objective composition. Hilbert spaces, complex amplitudes, gauge fixing, anomalies, and the Born rule require additional structure. The claim here is only that a local single-branch sharp generator, when it exists, carries response covectors and a Lagrangian endpoint relation.

Open Markov processes provide a close precedent. Baez, Fong, and Pollard show that open detailed-balanced Markov processes compose and that black-boxing yields Lagrangian relations between symplectic vector spaces [6]. The present construction starts instead from positive boundary-conditioned region weights and uses a sharp-generator limit to attach endpoint response covectors. Gauge theory and gravity supply a different lesson: cutting a gauge system may require additional boundary degrees of freedom to maintain gauge-invariant local phase spaces [11]. In the present language, such variables are candidates for boundary-completion data forced by gluing, although the full gauge-theoretic construction lies outside the finite-dimensional theorem.

6.3 limitations and extensions

The limitations are structural. Exact gluing may fail for the proposed boundary variables; a sharp generator may not exist; the generator may be nonsmooth or multibranched; and gauge, constrained, dissipative, or oscillatory systems may yield presymplectic, relation-level, or branch-dependent geometry rather than a global symplectic map. These cases are not exceptions to the claim, but boundaries of its hypotheses.

Natural extensions are suggested by those boundaries: Schrödinger bridges test whether endpoint dual potentials can be read as boundary responses; open Markov processes test the relation to black-boxed Lagrangian relations; constrained or gauge examples test boundary completion; and semiclassical path integrals test the distinction between exact amplitude composition and local stationary-phase response matching.

7 conclusion

Cut-objective predictive mechanics begins with bounded histories and their boundary-conditioned weights. Auxiliary internal cuts are allowed only when summing over complete data on the cut reproduces the weight of the whole region. In a positive sharp regime, the logarithmic generator composes by an infimum principle. At smooth interior minimizers, first-order variation gives the response balance $p_b^- - p_b^+ = -DI(b^*)$, reducing to matching of oriented boundary responses when the interface rate has zero derivative, including the subexponential case. For a C^2 finite-dimensional endpoint generator, the response relation is Lagrangian in $T^*\mathcal{M}^- \times T^*\mathcal{M}$, and under mixed-Hessian nondegeneracy it is a local symplectic map.

The result is a theorem-shaped bridge from composable history weights to phase geometry. It complements Hamilton-Jacobi theory and symplectic geometry by relocating the origin of their boundary variables. Action-like generators arise as sharp limits of composable predictive weights, and conjugate variables arise as boundary responses selected by sharp variational composition.

The distinction is operational. Starting from an action or endpoint generator returns familiar Hamilton-Jacobi and generating-function mechanics. Starting from a composable region weight and then taking a regular sharp limit explains why the same response geometry is attached to a predictive boundary.

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